

Nadine Claire Bradbury

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Education

University of California, Los Angeles (UCLA)

PhD, Chemistry, Theory and Computation

Oct 2019- Jun 2024

Advisor: Daniel Neuhauser

Thesis: Stochastic Electronic Structure Methods for Nano- to Microscale Molecular Complexes

California Institute of Technology (Caltech)

BSc, Chemistry

Oct 2015 - Jun 2019

Research Experience

Postdoctoral Research Associate

Sept 2024- Present

Princeton University

Princeton, NJ

- Postdoctoral work with Prof. Joseph Subotnik, on phase space electronic structure theory for to capture electron-nuclear correlations in solids and spin-nuclear correlation.

Graduate Student Research

Oct 2019 - June 2024

University of California, Los Angeles

Los Angeles, CA

- Novel stochastic electronic structure theory focused on improving the scaling and accuracy of Bethe-Salepeter equation methods.

Undergraduate Senior Thesis

Mar 2018 - Jun 2019

California Institute of Technology

Pasadena, CA

- Advisor: William A Goddard III
- Mechanistic insight into biased versus non-biased ligands for μ -opioid G-protein coupled receptors using molecular dynamics and drug docking softwares.

Caltech Japan Internship Exchange Intern

Jun 2018 - Aug 2018

Daiichi Corporation, Research and Development

Himeji, Hyogo, Japan

Teaching Assistant Experience

Chemical Thermodynamics

Spring 2021

Chem110a – Evaluations 8.1/9

UCLA

General Chemistry

Fall 2019 - Fall 2020

Chem20ab – Evaluations 8.0/9

UCLA

Nature of the Chemical Bond

Spring 2019

Ch120

Caltech

Research Presentations

Invited Talk – Sigtuna CISS Meeting

2026

Invited Talk – GRC Electron Spin Interactions with Chiral Molecules and Materials

2025

Contributed Presentation - ACS Fall 2023

2023

Contributed Presentation - ACS Spring 2021

2021

Awards & Honors

NSF Graduate Research Fellowship

2021-2024

George Gregory Excellence in Research Award

2023

UCLA Department of Chemistry and Biochemistry

Publications

Peer Reviewed Publications,

- JE Subotnik, E Alguire, N Bellonzi, X Bian, M Bhati, **NC Bradbury**, DV Cofer-Sabica, B Curlee, T Duston, S Fatehi, G Miao, Z Pei, L Peng, T Qiu, Z Tao, X Wu, Y Wu, Z Zaidi, Y Shao, J Rawlinson, N Shenvi, R Littlejohn, A Perspective on Phase Space Electronic Structure Theory : From Its Surface Hopping Origins Through To Its Future Promise, Submitted, (2026). [PDF]
- **NC Bradbury**, JE Subotnik, A Phase Space Electronic Structure View of The Solid State, Submitted, (2026).[PDF]
- M Bhati, DV Cofer-Shabica, JI Rawlinson, RG Littlejohn, JE Subotnik, **NC Bradbury**, Electronic Structure in a Phase Space, non-Born-Oppenheimer Framework: Geometric Forces and Moody-Shapere-Wilzcek Revisited, Submitted, (2026).[PDF]
- A Wright, **NC Bradbury**, D Neuhauser, EM Sletten, JR Caram, Exploring the Role of Excitonic Coupling in Polariton Formation, *J. Phys. Chem. Lett.* 17 (32): 9276–9284.(2026).[PDF]
- L Peng, T Duston, **NC Bradbury**, M Bhati, X Tao, M Rosen, JE Subotnik, A Conceptual Shift In Our Understanding of Degenerate Radical Spin Systems: Spin-Rotation Coupling Turned On Its Head, *J. Am. Chem. Soc.*, In Press (2026).[PDF]
- T Allen, BY Li, **NC Bradbury**, D Neuhauser, Environment-Induced Exciton Renormalization in the Photosystem II Reaction Center, *J. Chem. Theory Comput.*, 22, 10, 5115–5121, (2026).[PDF]
- J Williams, A Bailey, M Pengshung, AP Deshmukh, C Garcia, J Cao, BY Li, **NC Bradbury**, A Wright, C Chuang, D Neuhauser, E Sletten, JR Caram, Structural Engineering of Cyanine Dyes to Access Highly Redshifted and Emissive J-aggregates, *J. Am Chem. Soc.*, In Press (2026).[PDF]
- L Peng, T Qiu, **NC Bradbury**, X Bian, M Bhati, R Littlejohn, NM Kidwell, JE Subotnik, Phase Space Electronic Structure Theory: From Diatomic Lambda-Doubling to Macroscopic Einstein-de Haas, *J. Phys. Chem. Lett.* 17, 10, 2799–2811 (2026).[PDF]
- X Bian, T Duston, **NC Bradbury**, Z Tao, M Bhati, T Qiu, X Wu, Y Wu, JE Subotnik, The Phase-Space Way To Electronic Structure Theory and Subsequently Chemical Dynamics, *Chem. Phys. Rev.* 7, 011303 (2026). [PDF]
- T Duston, **NC Bradbury**, Z Tao, JE Subotnik, Conical Intersections and Electronic Momentum As Viewed From Phase Space Electronic Structure Theory, *J. Phys. Chem. Lett.* 16, 8994-9003 (2025).[PDF]
- **NC Bradbury**, T Duston, Z Tao, JI Rawlinson, R Littlejohn, JE Subotnik, Symmetry Breaking as Predicted by a Phase Space Hamiltonian with a Spin Coriolis Potential, *J Chem Phys*, 162, 244107 (2025). [PDF]

- BY Li, T Duong, T Allen, **NC Bradbury**, JR Caram, D Neuhauser, Parameterized Attenuated Exchange for Generalized TDHF@ v_W Applications, *J Chem Phys*, **163**, 034102 (2025).[PDF]
- Z Tao, T Qiu, X Bian, T Duston, **NC Bradbury**, JE Subotnik, A basis-free phase space electronic Hamiltonian that recovers beyond Born–Oppenheimer electronic momentum and current density, *J Chem Phys*, **162**, 144111 (2025).[PDF]

Prior to Princeton,

- **NC Bradbury**, BY Li, T Allen, JR Caram, D Neuhauser, No more gap-shifting: Stochastic many-body-theory based TDHF for accurate theory of polymethine cyanine dyes, *J. Chem. Phys.*, **161**, 141101 (2024). [PDF]
- **NC Bradbury**, RF Ribeiro, JR Caram, and D Neuhauser, Stochastic methodology shows molecular interactions protect two-dimensional polaritons, *Phys. Rev. B*, **109**, L241303 (2024).[PDF]
- M. Sereda, T. Allen, **N. C. Bradbury**, K. Z. Ibrahim, D. Neuhauser Sparse-Stochastic Fragmented Exchange for Large-Scale Hybrid TDDFT Calculations, *J. Chem. Theory Comp.*, **20**,10, 4196-4204 (2024) [PDF]
- **NC Bradbury**, T Allen, M Nguyen, and D Neuhauser, Deterministic/Fragmented-Stochastic Exchange for Large Scale Hybrid DFT Calculations, *J. Chem. Theory Comput.*, **19**, 24 (2023). [PDF]
- **NC Bradbury**, M Nguyen, K Ibrahim, and D Neuhauser, Optimized Attenuated Interaction: Enabling Stochastic Bethe-Salpeter Spectra for Large Systems, *J. Chem. Phys.*, **158**, 154104 (2023). [PDF]
- A Bailey, AP Deshmukh, **NC Bradbury**, M Pengshung, T Atallah, J Williams, U Barotov, D Neuhauser, EM Sletten, JR Caram, Exploring the Design of Superradiant J-Aggregates from Amphiphilic Monomer Units, *Nanoscale*, **15**, 3841-3849 (2023). [PDF]
- **NC Bradbury**, M Nguyen , JR Caram, and D Neuhauser, Bethe Salpeter Equation Spectra for Very Large Systems, *J. Chem. Phys.*, **157**, 031104 (2022). [PDF]
- AP Deshmukh, N Geue, **NC Bradbury**, C Chuang, M Pengshung, J Cao, EM Sletten, D Neuhauser, and JR Caram Bridging the gap between H- and J-aggregates: Classification and supramolecular tunability for excitonic band structures in two-dimensional molecular aggregates, *Chem. Phys. Rev.*, **3**, 021401 (2022). [PDF]
- **NC Bradbury**, C Chuang, AP Deshmukh, E Rabani, R Bear, JR Caram, and D Neuhauser, Stochastically Realized Observables for Excitonic Molecular Aggregates. *J. Phys. Chem. A*, **124**, 49, 10111-10120 (2020). [PDF] *Emily Carter Festshcrist Invited Paper*

Prior to UCLA,

- A Baranczak, Y Liu, S Connelly, WG Han Du, ER Greiner, JC Genereux, RL Wiseman, YS Eisele, **NC Bradbury**, J Dong, L Noodleman, KB Sharpless, IA Wilson, SE Encalada, and JW Kelly, A Fluorogenic Aryl Fluorosulfate for Intraorganellar Transthyretin Imaging in Living Cells and in *Caenorhabditis elegans*. *J. Am. Chem. Soc.*, **137**, **23**, 7404-7414 (2015). [PDF]
- S Connelly, **NC Bradbury**, IA Wilson, High Resolution Structure of Wild Type Human Transthyretin in Complex with 3,3',5,5'-tetrachloro-[1,1'-biphenyl]-4,4'-diol, *RCSB PDB*, **4MAS** (2013).[DOI]